Qi Wang

Engineering Postgraduate | Mechanical Modeling & Simulation Projects

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Profile

Master of Engineering student with a background in Transportation Equipment and Control Engineering. Project experience in SolidWorks part and assembly modeling, mechanism motion studies, and MATLAB/Simulink simulation. Completed internships involving product modeling, test-data organization and R&D support.

Education

The University of Sydney Aug 2026 – Present
Master of Engineering | Computational Engineering | Expected completion: Jan 2028

Southwest Jiaotong University Sep 2021 – Jun 2025
Bachelor of Engineering, Transportation Equipment and Control Engineering
Average: 83.84/100. Selected coursework: Machine Design (90/100), Engineering Test and Signal Analysis (91/100), Control Circuit Design and Simulation (EDA) (97/100), Automatic Control Principle.

Internship Experience

Wuhu Ruiji Intelligent Manufacturing Co., Ltd. Jul – Aug 2023
Research Assistant Intern

- Assisted with competitor research, product-test data organization and report preparation for R&D discussions.
- Supported component drafting and suspension torsion-bar design discussions; organized related design materials.

Sichuan Yu Suan Ni Technology Co., Ltd. Jul – Aug 2022
Product R&D Intern

- Used SolidWorks to model mechanical components from 2D drawings and existing CAD files, and revise models.
- Collaborated with the development team on model updates and Photoshop textures for project presentation.

Selected Engineering Projects

Semi-Active Suspension Control with CarSim–Simulink Dec 2024 – May 2025
Bachelor's thesis | MATLAB/Simulink, CarSim

- Developed quarter-car and half-car suspension models and configured a full-vehicle co-simulation interface.
- Compared passive suspension, PID, Skyhook, fuzzy PID and GA-tuned PID using acceleration RMS and related metrics.

Steel Plate Turning Mechanism Sep 2023 – Sep 2024
Mechanical design course project | SolidWorks, Motion

- Compared four mechanism concepts and modeled a dual-arm crank-rocker assembly and its components.
- Analyzed displacement, velocity, acceleration and drive torque; documented motion-study outputs.
- Performed preliminary motor and transmission sizing, mobility analysis and strength calculations.

Pantograph Guide-Plate Profile Design & Optimization Apr 2023 – May 2024
Undergraduate research training project (SRTP) | Project lead

- Coordinated tasks and documentation; contributed to MATLAB geometry and force analysis with ADAMS motion comparisons.
- Contributed to constrained profile optimization using a genetic algorithm; examined initial-point and spring-stiffness effects.

Tools & Languages

Mechanical modeling SolidWorks parts, assemblies and Motion studies through project work.

Simulation & fundamentals Project use of MATLAB/Simulink, CarSim and ADAMS; engineering drawing and control fundamentals.

Supporting tools & languages Excel data organization, Photoshop; Mandarin (native), English (CET-4).